

One Policy for the Whole Arm: Context-Conditioned Distillation for Safe Robotic Bed-Bathing

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Abstract—Bed-bathing is among the most requested and labor-intensive tasks in assistive care, and a robot that performs it must wipe a person’s skin *completely, systematically, and with clinically safe* contact forces. Reinforcement learning (RL) yields competent per-region wiping policies, but a separate network per body part is impractical to deploy, and the standard contact-counting reward optimizes the wrong objective—clearing targets aggressively while ignoring coverage and force safety. We contribute: (i) a single *Context-Conditioned Policy* (CCP) that takes a region identifier and a one-bit care level and replaces eight per-region specialists, matching or exceeding each and beating a flat multi-task policy by +33% targets cleared (12/12 region×seed pairs, sign test $p=0.0005$) while holding force in the therapeutic band 2–2.7× more often; (ii) a *Comfort reward* that shapes wiping along a self-supervised arm axis with a 2-D (length×circumference) coverage term; and (iii) a caregiver-selectable *gentle↔thorough* dial: flipping CCP’s care bit is a paired same-network comparison that raises therapeutic-band adherence by +34% (10/12 pairs, $p=0.039$) at an 8% coverage cost. Across a method ladder, learning dominates a scripted geometric sweep on every wiping-quality metric—the sweep traces near-perfect geometry yet fails sustained safe contact—so the reward design decides not how much the robot wipes but how safely.

Index Terms—Physically assistive devices, reinforcement learning, imitation learning, safety in HRI.

I. INTRODUCTION

Millions of people who cannot bathe independently rely on caregivers for bed-bathing, one of the most physically demanding and frequently requested activities of daily living. A robot that assists must move a soft tool across a person’s skin, covering each body region fully and in order while keeping contact force within a gentle, therapeutic range—an inherently *coverage-and-safety* problem in direct contact with a vulnerable human.

Reinforcement learning can train a competent wiping policy for a single arm region. Two obstacles keep that from being a system. First, **deployment**: a robot cannot carry a library of per-region networks and a switch to select among them. Second, **objective mismatch**: the reward shipped with public bed-bathing environments simply counts cleared targets, which encourages fast dabbling that inflates the count while leaving gaps and spiking force. A caregiver does the opposite—slow, systematic, gentle, sustained contact.

We address both with one idea: distill per-region, per-reward specialist policies into a single *Context-Conditioned Policy* (CCP) whose input carries the body region and a one-bit *care*

level, and train it from teachers optimized with a reward built for the true objective. Our contributions are:

- 1) **One policy for the whole arm.** A single context-conditioned network replaces eight region×reward specialists, matches or exceeds each in its own region, and beats a flat multi-task policy by +33% targets cleared (positive in all 12 region×seed pairs, exact sign test $p=0.0005$) while keeping force in-band 2–2.7× more often (Sec. VI-B).
- 2) **The Comfort reward.** A structured reward (Sec. IV-B) that shapes wiping along a self-supervised arm axis and adds a 2-D (length×circumference) coverage term from the same SVD’s second principal axis, producing complete, force-safe sweeps.
- 3) **A caregiver-selectable dial.** CCP’s care bit exposes a runtime gentle↔thorough trade-off (Fig. 1). Because it is one network, flipping the bit is a paired same-network test: the gentle setting raises therapeutic-band adherence by +34% (10/12 pairs, $p=0.039$) at an 8% coverage cost (Sec. VI-A).

Across a method ladder (Sec. VI-D), learning dominates a non-RL scripted sweep on every wiping-quality metric, and conditioned distillation dominates flat multi-task RL on both coverage and safety.

II. RELATED WORK

Assistive manipulation in contact with people. Simulation frameworks such as Assistive Gym [1] established RL benchmarks for bathing, dressing, and feeding with a human in the loop, where comfort and force are part of the objective. Prior bed-bathing and dressing work emphasizes safe, personalized contact and force regulation [2], [3]. Recent dressing policies treat force limits as an explicit constraint rather than a soft penalty [4]. We follow this line by making therapeutic-band force a first-class design and evaluation axis, and by exposing safety as a user-selectable behavior.

Policy distillation and conditioning. Distilling one or more RL teachers into a compact student is well established [5]; multi-task and goal-/context-conditioned policies let one network serve many settings. Our student is conditioned on both a body region and a care level, so a *single* network both generalizes across the arm and exposes a behavior dial—and

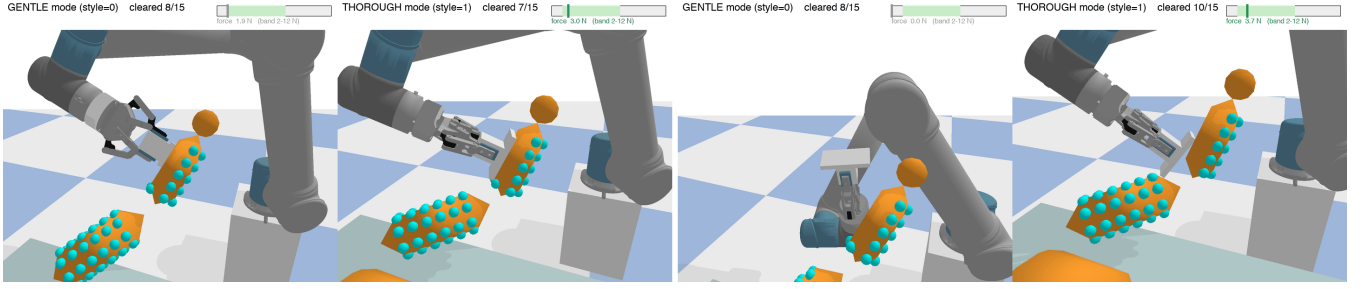


Fig. 1. **One policy, two behaviours.** The same Context-Conditioned Policy wiping a forearm at two time points (left block: early; right block: late), with its care bit set to *Gentle* (left of each block) and *Thorough* (right). The force gauge (top strip) stays in the therapeutic 2–12 N band; the cleared count rises as targets (cyan) are wiped. A caregiver selects the behaviour with a single bit—no retraining.

we show the conditioned student regularizes and can rescue weak teachers, rather than merely compressing them.

III. PROBLEM FORMULATION

A UR5 manipulator holds a soft tool and cleans a target region of a supine human arm in simulation (PyBullet). Each region carries up to 15 cleanable *targets*; a target is *cleared* when the tool passes within `clear_radius`=0.022 m while in contact. Episodes last 150 steps. We model the task as an MDP: the observation $o \in \mathbb{R}^{90}$ concatenates robot joint state, end-effector pose, tool–skin contact geometry, and a region/target encoding; the action $a \in \mathbb{R}^6$ is a continuous end-effector command. Safe contact is defined by a *therapeutic force band* of [2,12] N: below it the tool grazes without cleaning, above it risks discomfort or injury. The objective is to maximize coverage (targets cleared, systematic sweep) subject to keeping contact force in-band—a coverage-and-safety problem, not target collection.

IV. METHOD

A. Teacher policies

We train one PPO policy (RLlib 1.13) per (region, reward) pair. The policy/value network is `fcnet_hiddens=[100,100]` with `tanh`; `train_batch_size=19200`, `num_sgd_iter=50`, `sgd_minibatch_size=128`, trained in ~ 250 k-step chunks with best-checkpoint archiving.

B. Reward design

We compare two rewards on identical dynamics. The **Contact-count reward** R_{cc} is the baseline shipped with the public environment:

$$R_{cc} = 1.0(-d_{\text{arm}}) + 0.01(-\|a\|^2) + 5.0 c_t, \quad (1)$$

where d_{arm} is tool–arm distance and c_t the targets cleared at step t . It rewards *how many* targets are cleared but knows nothing of *where* the tool is along the arm or whether coverage is systematic.

The proposed **Comfort reward** R_{comf} keeps those base terms and adds four coverage-and-safety shaping terms (Table I):

$$R_{\text{comf}} = w_d r_{\text{dist}} + w_a r_{\text{act}} + w_c r_{\text{clr}} + w_s r_{\text{sweep}} + w_w r_{\text{win}} + w_e r_{\text{end}} + w_v r_{\text{cov}}. \quad (2)$$

A *two-phase distance* r_{dist} rewards approach while reaching and on-surface travel while wiping. A *self-supervised sweep axis* term uses the principal axis u of the target cloud (SVD, refreshed $\sim$ every 200 steps) and the tool’s normalized sweep coordinate $s = \text{clip}(\langle p - \bar{p}, u \rangle - s_{\min}) / \text{span}, 0, 1$; with $\Delta s_{\text{dir}} = \rho(s_t - s_{t-1})$ for pass direction $\rho \in \{\pm 1\}$,

$$r_{\text{sweep}} = \begin{cases} k_f \min(\Delta s_{\text{dir}}, \delta), & \Delta s_{\text{dir}} > \varepsilon \\ -k_b \min(-\Delta s_{\text{dir}}, \delta), & \Delta s_{\text{dir}} < -\varepsilon \\ 0, & \text{otherwise,} \end{cases} \quad (3)$$

active only in contact ($k_f=0.5, k_b=0.75$), rewarding monotone progress and mildly penalizing backtracking. A *local-window* term r_{win} credits contact/clears near the current s ; an *evidence-gated end-of-pass bonus* $r_{\text{end}}=4.0$ fires only when a pass reaches an end *and* accumulated real coverage evidence, preventing farming by sweeping empty space. Finally, the *2-D* (s, w) *coverage* term uses the *second* SVD axis—the circumferential (“column”) direction—to form a width coordinate w ; the first visit to each cell of a 4×4 (s, w) grid earns $r_{\text{cov}}=1$ once per episode. The *1-D* sweep terms reward progress *along* the arm; this term is what makes the policy cover *across* it (a grid search over $w_v \in \{0, 0.5, 1, 2\}$ chose $w_v=1$: 6.08 vs 5.68 cleared at $w_v=0$).

C. Context-Conditioned Policy (CCP)

CCP is a single diagonal-Gaussian MLP $\pi_{\theta}(a | o, c) = \mathcal{N}(\mu_{\theta}, \text{diag } \sigma_{\theta}^2)$ with a (128, 128) `tanh` backbone and heads for $\mu(6)$ and clamped $\log \sigma(6) \in [-5, 2]$. Its *context* c concatenates a region one-hot (4) with a one-bit *care level* $\kappa \in \{0, 1\}$. At deployment the one-hot selects the current body part and κ selects behavior: $\kappa=0$ (*Gentle*, Comfort-reward behavior) or $\kappa=1$ (*Thorough*, Contact-count behavior). One network covers the whole arm and both behaviors.

TABLE I
TERMS OF THE COMFORT REWARD R_{comf} (EQ. 2). THE CONTACT-COUNT
BASELINE R_{cc} USES ONLY THE FIRST THREE ROWS.

Term	Rewards	Weight
r_{dist}	on-surface proximity (two-phase)	1.0
r_{act}	$-\ a\ ^2$ (effort penalty)	0.01
r_{clr}	targets cleared this step c_t	5.0
r_{sweep}	monotone progress along arm axis	1.0
r_{win}	contact/clears near current s	1.0
r_{end}	evidence-gated end-of-pass bonus	1.0
r_{cov}	first visit to each (s, w) grid cell	1.0

Algorithm 1 Context-Conditioned Distillation

Require: teachers $\{\pi_r\}_{r \in R}$; context map $c(r)$; rollout length N ; weight β ; optional care values $\{\kappa_r\}$

- 1: $\mathcal{D} \leftarrow \emptyset$
- 2: **for** each region $r \in R$ **do**
- 3: roll out π_r for N steps; record $(o, \mu_r(o), \log \sigma_r(o), c(r))$
- 4: if $\{\kappa_r\}$ given, append κ_r to each $c(r)$
- 5: $\mathcal{D} \leftarrow \mathcal{D} \cup \mathcal{D}_r$
- 6: **end for**
- 7: train $\pi_\theta(a \mid o, c)$ by minibatch Adam on the mean of L (Eq. 4)
- 8: **return** single context-conditioned policy π_θ

D. Distillation

Each teacher π_r is rolled out in its region to record $(o, \mu_r(o), \log \sigma_r(o), c(r))$. Pooling all regions gives $\mathcal{D} = \bigcup_r \mathcal{D}_r$. CCP minimizes a behavior-cloning term on the mean plus a distribution-matching (KL) term on the full Gaussian:

$$L(\theta) = \|\mu_\theta - \mu_T\|^2 + \beta \text{KL}(\mathcal{N}_T \parallel \mathcal{N}_\theta), \quad (4)$$

where, per action dimension with log-std $\ell = \log \sigma$,

$$\text{KL}(\mathcal{N}_T \parallel \mathcal{N}_\theta) = \sum_{i=1}^6 \left[\ell_{\theta,i} - \ell_{T,i} + \frac{\sigma_{T,i}^2 + (\mu_{T,i} - \mu_{\theta,i})^2}{2\sigma_{\theta,i}^2} - \frac{1}{2} \right]. \quad (5)$$

We minimize the batch mean of Eq. 4 with Adam (lr 10^{-3} , $\beta=1$), matching both the teacher’s action mean and its uncertainty (Alg. 1). Appending the care bit to $c(r)$ yields the dial student; omitting it and pooling all teachers yields the *flat* student, an irreversible blend.

V. EXPERIMENTAL SETUP

Regions. Forearm back/front and upper-arm back/front (4). **Methods.** Three reward classes—Comfort RL, Contact-count RL, and a non-RL scripted geometric sweep—each as a per-region *specialist* and a distilled *student*; the proposed **CCP** distilled from all eight RL specialists and evaluated at both care settings; and a **flat** multi-task PPO baseline (Comfort reward, all regions, no conditioning). **Protocol.** Identical metric reducers, 200 episodes/region (100 for the deterministic scripted controller), force band $[2, 12]$ N, and **3 training seeds**

TABLE II
EVALUATION METRICS (EXACT DEFINITIONS).

Metric	Definition
Targets cleared	$\sum_t k_t$ (distinct, of 15)
In-band force frac.	$ \{t \in \mathcal{C} : f_t \in [2, 12]\} / \mathcal{C} $
Contact-time frac.	$ \mathcal{C} /T$
Sweep consistency	forward-progress wipe steps / wipe steps
Clear s -coverage	arm-axis bins with a clear, of 10
Clear spatial order	consecutive clears in dominant s -dir. (1.0=orderly)
Peak force	$\max_{t \in \mathcal{C}} f_t$ (worst case)

TABLE III
FLAGSHIP COMPARISON (GRAND MEAN, 4 REGIONS $\times$ 3 SEEDS).
BOLD=BEST IN COLUMN. CCP’S TWO CARE SETTINGS JOINTLY WIN
EVERY COLUMN; BASELINES ARE DOMINATED.

Method	Coverage / systematic			Safety / engag.	
	Clr/15	Sweep	s -cov	In-band	Cont.-t
Comfort spec.	5.79	0.184	0.336	0.173	0.571
Contact-cnt spec.	6.38	0.195	0.354	0.154	0.554
Student $\leftarrow$ Comfort	6.27	0.184	0.342	0.193	0.607
Student $\leftarrow$ Contact-cnt	6.68	0.201	0.351	0.153	0.590
CCP @Gentle	6.17	0.172	0.343	0.197	0.612
CCP @Thorough	6.80	0.206	0.355	0.156	0.592
Flat (multi-task)	5.11	0.181	0.286	0.074	0.417
Scripted (non-RL)	4.56	0.046	0.260	0.039	0.339

per learned method. For significance we use exact sign tests over the 12 region $\times$ seed pairs plus a conservative per-seed paired t -test—the seed, not the episode, is the statistical unit. Every number is emitted by `analyze_final.py` from the metrics CSV.

Metrics. Per step t an episode yields force f_t , an in-contact flag, sweep coordinate s_t , per-step clears k_t ; $\mathcal{C} = \{t : \text{contact}\}$ and T the episode length. Table II defines the reported set. Two families are excluded from cross-method comparison: *pass completions* (counts the Comfort reward’s own gate—one method’s machinery) and *mean/p95 force* (reward *not touching*: the scripted sweep has the lowest mean force because it grazes), so force safety is reported as in-band fraction.

VI. RESULTS

Table III is the flagship comparison: grand means over 4 regions $\times$ 3 seeds, with the best value per column in bold. The two CCP settings capture every column—Thorough wins the coverage metrics, Gentle wins the safety/engagement metrics—and both dominate the flat and scripted baselines. Fig. 2 plots the same data as a safety $\leftrightarrow$ coverage frontier: the comfort dial moves CCP along it while the baselines sit strictly inside.

A. A caregiver-selectable gentle $\leftrightarrow$ thorough dial

CCP spans the trade-off at deployment time. Averaged over 4 regions $\times$ 3 seeds: Thorough mode 6.80 cleared / 0.156 in-band; Gentle mode 6.17 cleared / 0.198 in-band,

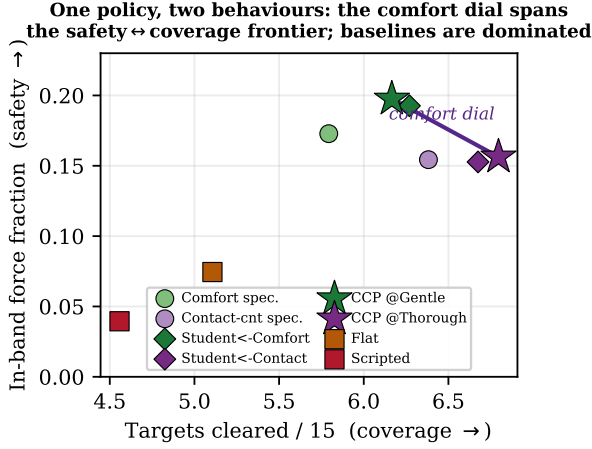


Fig. 2. The safety↔coverage frontier. Each point is a method (grand mean). Flipping CCP’s one care bit slides it from Thorough (best coverage) to Gentle (best in-band force) along the frontier; the flat and scripted baselines lie strictly inside.

TABLE IV
THE COMFORT DIAL PER REGION (3-SEED MEANS). GENTLE IS GENTLER (HIGHER IN-BAND) IN 4/4 REGIONS; THOROUGH CLEARS $\geq$ IN 4/4.

Region	Gentle		Thorough	
	clr	/ in-band	clr	/ in-band
forearm back	6.35	/ 0.168	6.95	/ 0.121
forearm front	6.15	/ 0.311	7.65	/ 0.278
upper-arm back	6.62	/ 0.165	7.02	/ 0.106
upper-arm front	5.55	/ 0.146	5.56	/ 0.120

with the ordering consistent on all four regions (Table IV). Because CCP is *one* network, flipping κ is a perfectly paired comparison—same weights, seeds, regions; only the care bit differs. Switching to Gentle raises **in-band force fraction** by +34% (better in 10/12 region×seed pairs, exact sign test $p=0.039$) at an 8% coverage cost (3/12, n.s.). On composite *safe clears* (cleared×in-band), the Comfort-distilled behavior leads by +28%. The raw clear-count ordering is reversed (Thorough clears more): the reward design does not make wiping more *productive*, it makes it *safer*, and the dial hands that choice to the caregiver.

B. CCP beats the flat multi-task baseline

In Thorough mode CCP clears **6.80** vs **5.11** for the flat baseline—+33%, positive in 12/12 **region×seed pairs** (exact sign test $p=0.0005$), consistent per seed (6.43/6.81/7.14 vs 5.26/5.37/4.69). The Comfort-distilled student alone gains +1.16 clears/seed (per-seed paired $t=3.76$, $n=3$). Conditioning also buys safety: every conditioned student keeps force in-band $2\text{--}2.7\times$ more than the flat baseline (0.153–0.198 vs 0.074). One network replaces four specialists and beats the unconditioned alternative on both axes (Fig. 3).

C. Distillation $\geq$ specialists—and rescues weak teachers

CCP-family students match specialists where those are strong and repair them where weak (cleared/15): Comfort

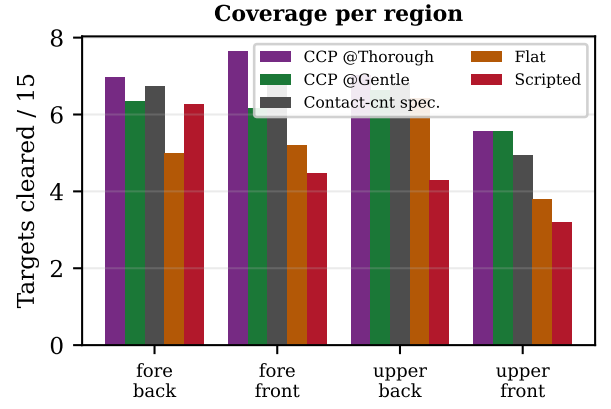


Fig. 3. Targets cleared per region. CCP (both settings) exceeds the flat and scripted baselines in every region; the residual gap on *back* regions is a physical reachability limit (Sec. VII), not a policy failure.

teacher→student 4.99 → **6.23** on forearm front and 5.19 → **5.96** on upper-arm front; Contact-count 4.95 → **5.69** on upper-arm front. No region regresses by more than 0.3. Grand means: students 6.17–6.80 vs specialists 5.79–6.38—pooling regularizes rather than merely compressing.

D. Ablating the design: learning and conditioning both matter

The rows of Table III form an ablation of the full method. Removing the coverage term ($w_v=0$) drops cleared from 6.08 to 5.68. Removing region conditioning yields the *flat* baseline (5.11 / 0.074). Removing learning entirely yields the *scripted* sweep (4.56 / 0.039). Removing the care bit yields the single-family students, which cannot switch behavior. The result is a monotone ladder: scripted < flat < specialists (5.79–6.38) < CCP students (6.17–6.80). The scripted controller is not merely less productive—it fails the wiping-*quality* battery, in contact only 25–41% of steps (vs 51–74% for CCP) with in-band force collapsing to 0.02 on the upper arm—yet its near-perfect clear spatial order (0.997) shows it traces textbook geometry. Raw clear-count and geometric orderliness are mirages of wiping; sustained safe contact had to be learned. Crucially, **distillation cannot manufacture it**: cloning the scripted controller into a student adds a little coverage (4.56 → 5.05) but leaves in-band force at ~ 0.04 vs ~ 0.19 for RL students. Distillation *amplifies* what a teacher already does; the generalization is a property of the learned behavior, not of the distillation mechanism.

VII. LIMITATIONS

Reachability. One angular column of each *back* region faces the bed—the supine limb’s support surface. Per-target IK over a dense orientation grid, dynamic clearability under three controllers, and base-/posture-pose sweeps all show no configuration of this environment can wipe it (the humanoid lacks a forearm-pronation DOF). This caps targets-cleared near 10/15 on back regions and keeps *task_success* (69% threshold) near zero for *all* methods; the support-facing surface

requires physically re-positioning the limb, a caregiver action assigned to the pipeline’s classical phases. **Force spikes.** Peak contact forces reach 47–68 N for all methods; bounding worst-case force (force-rate shaping or a controller-level clamp) is required before hardware. **Statistics.** Sign tests over 12 pairs are exact, but per-seed $n=3$; gaps quoted without a p -value are directional. **Scope.** Results are simulation-only; sim-to-real transfer is future work.

VIII. CONCLUSION

A single Context-Conditioned Policy, conditioned on body region and a one-bit care level, wipes all four arm regions, beats a flat multi-task baseline in every region $\times$ seed pair (+33%, $p=0.0005$), and exposes a caregiver-selectable gentle $\leftrightarrow$ thorough dial whose gentle end raises therapeutic-band adherence by 34% in a paired same-network test ($p=0.039$). Learning dominates scripting on every wiping-quality metric, and conditioned distillation dominates flat multi-task RL on both coverage and safety. The reward design decides not how *much* a robot wipes but how *safely*; distillation turns that choice into a runtime-switchable property of one deployable network. Bounding peak force and sim-to-real transfer are the next steps toward hardware trials.

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